

Sai Monica R

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Summary

Associate AI Software Developer with 1+ years of experience and an Integrated M.Tech in Computer Science and Engineering (Data Science), seeking Junior or Associate Data Scientist and AI/ML Engineer roles. Skilled in data preprocessing, exploratory data analysis, feature engineering, and building and evaluating machine learning models with scikit-learn and PyTorch. Experienced in developing LLM applications with RAG, tool-calling agents, and evaluation suites, and in deploying models as REST APIs with FastAPI, Docker, Kubernetes, and CI/CD.

Education

Vellore Institute of Technology

Integrated M.Tech in Computer Science and Engineering (Data Science) — CGPA: 8.63

Sep 2020 – Jun 2025

Vellore, India

Skills

Languages: Python, SQL, JavaScript, Java

Data Science & ML: Pandas, NumPy, scikit-learn, PyTorch, EDA, Feature Engineering, Model Evaluation, Tableau

Generative AI: LLM APIs (OpenAI, Gemini, Groq), LangChain, Hugging Face, Prompt Engineering, RAG, MCP

Backend & Databases: FastAPI, Flask, Node.js, Express.js, React.js, REST APIs, PostgreSQL, MySQL

DevOps & Testing: Docker, Kubernetes, Jenkins, CI/CD, Git, Selenium, Unit and Integration Testing

Experience

ESDS Software Solutions

Jul 2025 – Present

Associate AI Software Developer

Chennai, India

- Developed and maintained full-stack enterprise applications with React.js front ends, Node.js/Express.js and FastAPI back ends, and PostgreSQL databases, writing testable, production-quality code
- Built, deployed, and maintained 30+ REST API endpoints and microservices, designing endpoints and database interactions to support business workflows
- Integrated LLM-based features into production systems, applying prompt engineering and evaluation to improve the quality and consistency of model responses
- Debugged and performed root cause analysis on model and application issues, tracing problems across the front end, API, and database layers
- Automated build, test, and deployment on every code commit using CI/CD pipelines with Docker and Kubernetes, improving deployment reliability
- Adopted Claude Code as an agentic coding assistant with a review-before-merge workflow, delivering features about 60% faster and reducing rework by 40%
- Collaborated with cross-functional teams to translate business requirements into technical solutions, following Git-based code review workflows

Nokia

Aug 2024 – May 2025

Software Development Intern

Bengaluru, India

- Developed an automated end-to-end UI testing framework using Python, Selenium, and Robot Framework, structured with the Page Object Model (POM) for reusable, maintainable test code
- Integrated automated test suites into Jenkins CI/CD pipelines, improving test coverage by 45% and reducing manual testing effort by 40%
- Automated test workflows and deployment processes using shell scripting in WSL environments

Ashok Leyland

May 2023 – Jun 2023

AI Intern

Chennai, India

- Built an LLM-powered document chatbot using the OpenAI API and LangChain, applying retrieval-augmented generation (RAG) to answer questions from internal documents
- Generated embeddings for document chunks and stored them in a FAISS vector database, enabling semantic retrieval of the most relevant context for each query
- Developed a Streamlit interface for the chatbot, reducing manual document search effort and making information easier to access

Projects

AI Voice Assistant (Agentic LLM App) — Python, FastAPI, Groq LLM

Apr 2026 – Jul 2026

- Built an end-to-end voice assistant that converts speech to text with the Google Web Speech API, reasons with a Groq-hosted LLM (Llama 3.3 70B), and replies by voice using the ElevenLabs Text-to-Speech API
- Served the assistant through a FastAPI WebSocket API for real-time, two-way communication
- Designed a multi-step ReAct tool-calling agent that decides which of 15 tools to use, covering web search, news, weather, and system control
- Integrated Google Calendar through OAuth 2.0 (Google API Python Client) so users can create and manage meetings and events by voice
- Built an LLM-as-judge evaluation suite that scores tool-selection accuracy and response quality against a baseline, catching regressions after prompt and tool changes

Disease Prediction using Random Forest — Python, scikit-learn, Flask

May 2022 – Jul 2022

- Preprocessed a medical symptom dataset and performed feature selection across 132 symptoms to prepare model inputs
- Trained and evaluated a scikit-learn Random Forest classifier that achieved 85% accuracy in predicting diseases from symptoms
- Deployed the model as a Flask REST API for real-time predictions, using Pickle to load the trained model efficiently at inference time

Data Science Salaries Dashboard — Python, Tableau

May 2025 – Jun 2025

- Performed exploratory data analysis (EDA) on a global data science salaries dataset to identify patterns and trends
- Built an interactive Tableau dashboard showing salary distribution, geographic trends, and differences across job roles

Certifications

AI Fluency (Anthropic) | AI-900: Azure AI Fundamentals (Microsoft) | AI Engineer Agentic Track: Agents & MCP (Udemy)